Customer Order Management System

SQL Project Documentation

Objective

A relational database for customers, products, orders and order line items. The project demonstrates PK, FK, NOT NULL, UNIQUE and CHECK constraints, order CRUD operations, total calculation, and customer purchase analysis.

1. Orders & Order_Details Table Design

Table	Column	Purpose / Constraint
Customer	Customer_ID	Primary key
Customer	Customer_Name	NOT NULL
Customer	Email / Phone	NOT NULL, UNIQUE
Product	Product_ID	Primary key
Product	Product_Name	NOT NULL
Product	Unit_Price	NOT NULL, CHECK >= 0
Orders	Order_ID	Primary key
Orders	Customer_ID	FK → Customer; NOT NULL
Orders	Order_Date	NOT NULL; defaults to current date
Orders	Total_Amount	NOT NULL, CHECK >= 0
Orders	Order_Status	CHECK: Pending, Shipped, Delivered, Cancelled
Order_Details	Order_Detail_ID	Primary key
Order_Details	Order_ID	FK → Orders; ON DELETE CASCADE
Order_Details	Product_ID	FK → Product
Order_Details	Quantity	NOT NULL, CHECK > 0
Order_Details	Price	NOT NULL, CHECK >= 0
Order_Details	Order_ID + Product_ID	UNIQUE together

2. Updated ER Diagram

Relationships: Customer 1:N Orders; Orders 1:N Order_Details; Product 1:N Order_Details.

CUSTOMER (PK Customer_ID) ■■1:N■■ ORDERS (PK Order_ID, FK Customer_ID)

ORDERS ■■1:N■■ ORDER_DETAILS (PK Order_Detail_ID, FK Order_ID, FK Product_ID) ■■N:1■■ PRODUCT (PK Product_ID)

3. Sample Customer, Order & Order_Details Data

Customers

ID	Customer Name	Email	Phone	Address
1	Arun Kumar	arun.kumar@example.com	9000000001	Tirunelveli
2	Priya Devi	priya.devi@example.com	9000000002	Chennai
3	Karthik Raj	karthik.raj@example.com	9000000003	Coimbatore
4	Meena S	meena.s@example.com	9000000004	Madurai

Orders

Order ID	Customer ID	Date	Total	Status
1001	1	2026-10-01	1200.00	Delivered
1002	2	2026-10-03	2100.00	Shipped
1003	1	2026-10-08	1200.00	Pending
1004	3	2026-10-10	1800.00	Delivered
1005	4	2026-10-12	900.00	Pending
1006	2	2026-10-15	2400.00	Delivered

Order_Details

Detail ID	Order ID	Product ID	Quantity	Price
1	1001	103	1	1200.00
2	1002	101	2	750.00
3	1002	102	1	600.00
4	1003	104	2	600.00
5	1004	105	1	1800.00
6	1005	102	2	450.00
7	1006	103	2	1200.00

4. Customer Order History & Purchase Analysis Report

The SQL script provides queries for customer order history, product-wise orders, number of times each product was ordered, total quantity sold, customer(s) with the maximum number of orders, total spending by customer, average order value and total sales. Sales and spending reports exclude cancelled orders.

Report	What it shows
Customer order history	Customer, order date, status and total
Product-wise report	Order count, quantity sold and line sales per product
Times ordered	Count of distinct orders per product
Total quantity sold	Sum of quantities in non-cancelled orders
Maximum orders	Customer(s) tied for the largest order count
Customer spending	Sum of non-cancelled order totals by customer
Average order value	Average non-cancelled order total
Total sales	Sum of non-cancelled order totals

5. Query Output Screenshots

Run the SQL script in MySQL Workbench and capture actual result-grid screenshots. Suggested screenshots: customer/product tables; Orders and Order_Details rows; complete joined order details; customer order history; product-wise report; total quantity sold; customer spending; average order value and total sales. Actual execution screenshots are not included because they must come from your running database.

SQL Script Contents

The accompanying .sql file includes database/table creation, constraints, sample INSERT statements, create-order and multi-product operations, total calculation, status and quantity updates, cancelled-order deletion, complete order JOIN, and all requested sales-analysis queries.

Execution note: the SQL starts with DROP DATABASE IF EXISTS CustomerOrderDB, which deletes an existing database of that name. Back up existing data before execution. The script includes demonstration UPDATE and DELETE statements; run report queries before those statements if you want to preserve the original sample data.